

Wanyu Bian

📍 San Francisco, CA USA

✉ wanyubian6@gmail.com

☎ 3527459592

in [LinkedIn](#)

🎓 [Google Scholar](#)

🐙 [Github](#)

SUMMARY

Machine Learning Scientist in Applied Mathematics with 7 years research experiences and 10 first author published papers in topics of: Machine Learning Optimization Algorithms ◦ Gen AI ◦ Diffusion model ◦ Ensemble Learning ◦ Computer Vision ◦ Inverse Problems

EDUCATION

University of Florida

Ph.D. in Applied Mathematics, advisor: Yunmei Chen

Gainesville, FL

08/2017 - 08/2022

University of Florida

M.S. in Applied Mathematics, advisor: Yunmei Chen

Gainesville, FL

08/2017 - 05/2019

University of Missouri

B.S. in Applied Mathematics & B.S. in Statistics

Columbia, MO

08/2013 - 05/2017

WORK EXPERIENCE

Machine Learning Scientist | Neuro42

10/2023 - present

- Develop and maintain generative AI/ML models for medical image clinical diagnostics, anomaly/tumor detection, tissue segmentation, EMI noise suppression, image reconstruction, magnets array arrangement optimization.

Research Fellow | Harvard Medical School

05/2022 - 10/2023

- **Multi-Modality training** algorithm using Meta-learning.
- Motion-compensated object tracking using **Vision Transformer**.
- **Diffusion model and self-supervised learning** on native data domain.

Graduate Assistant | Department of Math, University of Florida

08/2017 - 05/2022

- Research development on optimization-based Meta-Learning model, **ensemble multi-task learning**, solve inverse problems.
- Taught and tutored classes for Numerical Linear Algebra, Numerical Analysis, Calculus, etc.

TECHNICAL SKILLS & LANGUAGES

- **Programming Skills** – Python, Matlab, SQL, SAS, R
- **Frameworks and Platforms** – Pytorch, Tensorflow, Keras, Scikit-learn, Colab, Git, AWS

PUBLICATIONS

1. F. Liu, L. Zhang, **W. Bian**, X. He, "Chapter 29 Emerging AI Technology in MRI: Reconstruction, ImageProcessing, Analysis, and Beyond," *Clinical MR Physics: State-of-the-Art Practice*, AAPM Monograph No.42 ISBN: 9781951134358 [\[link\]](#)
2. **W. Bian**, A. Jang, F. Liu, "**Multi-task** Magnetic Resonance Imaging Reconstruction using Meta-learning," *Magnetic Resonance Imaging*, 116 (2025): 110278. [\[link\]](#)
3. **W. Bian**, A. Jang, L. Zhang, X. Yang, Z. Stewart, F. Liu, "**Diffusion Modeling** with Domain-conditioned Prior Guidance for Accelerated MRI and qMRI Reconstruction," *IEEE Transactions on Medical Imaging*, Aug 2024. [\[link\]](#)
4. **W. Bian**, Y. K. Tamilselvam, "A Review of Optimization-Based Deep Learning Models for MRI Reconstruction," *Applied Math*, 4.3 (2024): 1098-1127. [\[link\]](#)
5. **W. Bian** et al., "A Review of Electromagnetic Elimination Methods for Low-field Portable MRI Scanner," *IEEE ICMLCA2024*. [\[link\]](#)
6. **W. Bian**, "A Brief Overview of Optimization-Based Algorithms for MRI Reconstruction Using Deep Learning," *Preprint*, June 2024. [\[link\]](#)
7. **W. Bian**, A. Jang, F. Liu, "Accelerating Quantitative MRI using **Self-supervised** Deep Learning with Model Reinforcement," *Proceedings of the 32nd Annual Meeting of ISMRM*, May 2024. (**Oral presentation**) [\[link\]](#)
8. **W. Bian**, A. Jang, F. Liu, "qDiMo: Domain-conditioned Diffusion Modeling for Accelerated qMRI Reconstruction," *Proceedings of the 32nd Annual Meeting of ISMRM*, May 2024. (**Oral presentation**) [\[link\]](#)
9. **W. Bian**, A. Jang, F. Liu, "Improving Quantitative MRI Using Self-supervised Deep Learning with Model Reinforcement: Demonstration for Rapid T1 Mapping," *Magnetic Resonance in Medicine*, Feb 2024. [\[link\]](#)
10. **W. Bian**, A. Jang, F. Liu, "Magnetic Resonance Parameter Mapping Using Self-supervised Deep Learning with Model Reinforcement," *Preprint*, July 2023. [\[link\]](#)
11. **W. Bian**, "Optimization-Based Deep Learning Methods for Magnetic Resonance Imaging Reconstruction and Synthesis," *Department of Mathematics, University of Florida*, May 2022. [\[link\]](#)
12. **W. Bian**, Q. Zhang, X. Ye, Y. Chen, "A Learnable Variational Model for Joint Multi-modal MRI Reconstruction and Synthesis," *Provisional Acceptance by MICCAI 2022*. [\[link\]](#)
13. **W. Bian**, Y. Chen, X. Ye, "An **Optimal Control** Framework for Joint-channel Parallel MRI Reconstruction Without Coil Sensitivities," *Magnetic Resonance Imaging*, Volume 89, 2022, Pages 1-11. [\[link\]](#)
14. **W. Bian**, Y. Chen, X. Ye, Q. Zhang, "An Optimization-Based **Meta-Learning** Model for MRI Reconstruction with Diverse Dataset," *Journal of Imaging* 7, no. 11: 231, 2021. [\[link\]](#)
15. **W. Bian**, Y. Chen, X. Ye, "Deep Parallel MRI Reconstruction Network Without Coil Sensitivities," *The 3rd International Workshop on*

Patent Pending

- Machine Learning System For Generating Context-Aware Expressive Animations With Emotional Intelligence | Application # 19/242,890 | Filed: Jun 2025. | **Role: Inventor**
- Vision-Language Artificial Intelligence System for Automated Medical Diagnostic Indication with Multimodal Knowledge Integration | Application # 19/240,016 | Filed: Jun 2025. | **Role: Inventor**
- Methods and Systems for Ultra-Lowfield Magnetic Resonance Image Denoising Using Self-Supervised Learning | Application # 63/741,377 | Filed: Nov 2024. | **Role: Lead Inventor**
- Methods and Systems for Optimizing Magnet Array Configurations | Application # 63/740,816 | Filed: Nov 2024. | **Role: Lead Inventor**
- Methods For Noise Reduction On Measurement Domain | Application # 63/750,253 | Filed: Nov 2024. | **Role: Lead Inventor**
- Methods and Systems for Correction of Asymmetric Noise Pattern in Low-Field MRI | Application # 63/741,764 | Filed: Nov 2024. | **Role: Contributing Inventor**

TALKS & PRESENTATIONS

- qDiMo: Domain-conditioned Diffusion Modeling for Accelerated qMRI Reconstruction, at ISMRM 2024, May 2024. (Oral presentation) [\[link\]](#)
- Accelerating Quantitative MRI using Self-supervised Deep Learning with Model Reinforcement, at ISMRM 2024, May 2024. (Oral presentation) [\[link\]](#)
- Accelerated qMRI for Tissue Quantification using Generative AI and Diffusion Modeling, at Harvard Medical School, Oct 2023. (Oral presentation) [\[link\]](#)
- A learnable variational model for joint multimodal mri reconstruction and synthesis, at MICCAI 22, Sep 2022.
- An Optimization-Based Meta-Learning Model for MRI Reconstruction with Diverse Dataset, at SIAM Conference on Imaging Science (IS22), held virtually, March 2022.
- Learnable Optimization Algorithm for MRI Reconstruction with Meta-learning Model, at University of Florida Chapter of SIAM (SIAM Gators), held virtually, Oct 2021.
- Deep Parallel MRI Reconstruction Network Without Coil Sensitivities, at MICCAI-MLMIR'20, held virtually, Oct 8 2020. (MICCAI Student Participation Award)
- Deep Neural Network for Parallel MRI Reconstruction Without Coil Sensitivity Maps, at SIAM Conference on Imaging Science (IS20), held virtually, July 16 2020. (SIAM Student Travel Award)

AWARDS & LEADERSHIP

- MICCAI 2022 NIH PARTICIPATION AWARDS, MICCAI 2022 Sep 2022
- President of Graduate Student Chapter of the AMS at University of Florida Jan 2020 - May 2022
- Research Chair of Association for Women in Mathematics for UF Chapter Sep 2019 - May 2022
- MICCAI Student Participation Award, MICCAI 2020 Oct 2020
- SIAM Student Travel Award to attend the 2020 SIAM Conference on Imaging Science July 2020
- Winner of Best in show, DataFest 2017 April 2017
- Scholarship in College of Art & Science, University of Missouri Fall 2015 and Spring 2016

PROJECTS

- Brain MRI Diagnostic Indication Generation [\[link\]](#) | neuro42 12/2024 - current
 - Developed a language model diagnostic system for brain MRI analysis, leveraging **Vision Transformer (ViT)** and **Transformer Decoder models** to automatically generate clinical indications from MRI scans.
 - BLEU score=0.82. This measures similarity between generated indications and radiologist reports (higher is better, max = 1.0).
- Self-supervised learning model for T1-weighted contrast enhancement [\[link\]](#) | neuro42 05/2024 - 08/2024
 - Developed an image contrast enhancement model to enhance the contrast of gray matter/white matter.
- Self-supervised learning model for EMI noise suppression [\[link\]](#) | neuro42 11/2023 - 12/2023
 - Developed **Self-supervised Learning Noise Reduction Model** for EMI suppression.
 - SNR improved 11 dB in average and comparable to the data from high field scanner.
- Diffusion model for reconstruction [\[link\]](#) | Harvard Medical School 06/2023 - 10/2023
 - Developed **diffusion model** that conditioned on native data domain. The prior physics information are used as embeddings in the diffusion model. Reconstructed PSNR improved 2 dB comparing to state-of-the-art.
- Multi-task MRI Reconstruction using Meta-learning | Harvard Medical School 12/2022 - 03/2023
 - Conducted **multi-task learning** for reconstructing multiple sequences of images scanned in different contrasts using **Meta-Learning**.
 - Implemented a multi-task learning model, achieving a reduction in processing time by 86%. The model removes aliasing artifacts and preserve tissue details compared to individual training with highly under-sampling ratio (only 1/6 of signal was scanned).

Professional Service

- Journal/Conference Reviewer** for AAAI-26, ICML 2025, ICLR 2025, NeurIPS 2024&2025, MICCAI 2024, IEEE Transactions on Medical Imaging, Journal of Digital Imaging, Journal of Imaging Informatics in Medicine, MDPI AppliedMath and MDPI Mathematics.